

# Route Statistics API — Request & Response Reference

All endpoints are `POST` , base URL `http://localhost:4000` .

All bodies are JSON ( `Content-Type: application/json` ).

No auth required (portal-only endpoints).

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## Common Response Envelope

Every route-stats endpoint returns the same envelope — only `metric` , `scope` / `scopeValue` , and `value` differ.

```
{
  "success": true,
  "scope": "session" | "dateRange",
  "scopeValue": <see per-endpoint>,
  "metric": "<metricName>",
  "value": <number | object>
}
```

**Error (all endpoints):**

```
{ "error": "Failed to compute statistics", "message": "..." }
```

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## Session-Scoped Endpoints

**Parameter for all:** `sessionId` (number, required) in the JSON body.

Valid pre-seeded session IDs: `18429` , `57243` , `90618` , `34790` , `62851`

### 1. Total Routes Completed

`POST /api/route-stats/total-routes-completed-by-session`  
Body: `{ "sessionId": 18429 }`

```
{
  "success": true,
  "scope": "session",
  "scopeValue": 18429,
  "metric": "totalRoutesCompleted",
  "value": 42
}
```

`value` : `number` — count of completed route cycles for that session.

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### 2. Total Connected Users

`POST /api/route-stats/total-connected-users-by-session`

Body: { "sessionId": 18429 }

```
{
  "success": true,
  "scope": "session",
  "scopeValue": 18429,
  "metric": "totalConnectedUsers",
  "value": 15
}
```

value : number — count of `PlayerSession` rows for that session.

---

### 3. Users Started More Than One Route

POST /api/route-stats/users-started-more-than-one-route-by-session  
Body: { "sessionId": 18429 }

```
{
  "success": true,
  "scope": "session",
  "scopeValue": 18429,
  "metric": "usersStartedMoreThanOneRoute",
  "value": 7
}
```

value : number — count of players who started 2+ routes.

---

### 4. Users Completed More Than One Route

POST /api/route-stats/users-completed-more-than-one-route-by-session  
Body: { "sessionId": 18429 }

```
{
  "success": true,
  "scope": "session",
  "scopeValue": 18429,
  "metric": "usersCompletedMoreThanOneRoute",
  "value": 5
}
```

value : number — count of players who completed 2+ routes.

---

### 5. Users Never Started a Route

POST /api/route-stats/users-never-started-route-by-session  
Body: { "sessionId": 18429 }

```
{
  "success": true,
  "scope": "session",
  "scopeValue": 18429,
  "metric": "usersNeverStartedRoute",
  "value": 3
}
```

value : number — count of players who never started any route.

---

## 6. Users Started But Not Finished

```
POST /api/route-stats/users-started-but-not-finished-by-session
Body: { "sessionId": 18429 }
```

```
{
  "success": true,
  "scope": "session",
  "scopeValue": 18429,
  "metric": "usersStartedButNotFinished",
  "value": 2
}
```

value : number — count of players who started at least one route but never completed any.

---

## 7. Total Replays

```
POST /api/route-stats/total-replays-by-session
Body: { "sessionId": 18429 }
```

```
{
  "success": true,
  "scope": "session",
  "scopeValue": 18429,
  "metric": "totalReplays",
  "value": 10
}
```

value : number — count of route cycles where cycleNumber > 1 .

---

## 8. Avg Routes Per User

```
POST /api/route-stats/avg-routes-per-user-by-session
Body: { "sessionId": 18429 }
```

```
{
  "success": true,
  "scope": "session",
  "scopeValue": 18429,
  "metric": "avgRoutesPerUser",
  "value": 2.75
}
```

value : number — float rounded to 2 decimal places. 0 if no users.

---

## 9. Avg Routes Per Session

POST /api/route-stats/avg-routes-per-session-by-session  
Body: { "sessionId": 18429 }

```
{
  "success": true,
  "scope": "session",
  "scopeValue": 18429,
  "metric": "avgRoutesPerSession",
  "value": 8.5
}
```

value : number — float rounded to 2 decimal places.

---

## 10. Replay Ratio

POST /api/route-stats/replay-ratio-by-session  
Body: { "sessionId": 18429 }

```
{
  "success": true,
  "scope": "session",
  "scopeValue": 18429,
  "metric": "replayRatio",
  "value": {
    "byPlayer": 0.4667,
    "byRoute": 0.1923
  }
}
```

value : **object** — the only non-scalar value across all route-stats endpoints.

- byPlayer : fraction of players who replayed at least once (4 decimal places)
  - byRoute : fraction of all route cycles that were replays (4 decimal places)
- 

## Date-Range-Scoped Endpoints

Parameters for all:

| Field     | Type   | Required | Notes                                                                                                           |
|-----------|--------|----------|-----------------------------------------------------------------------------------------------------------------|
| startDate | string | yes      | ISO 8601 — YYYY-MM-DD                                                                                           |
| endDate   | string | no       | ISO 8601 — YYYY-MM-DD. If omitted, query runs from startDate to present. End is normalized to 23:59:59.999 UTC. |

scopeValue in response:

```
// with endDate
"scopeValue": { "startDate": "2025-01-01", "endDate": "2025-12-31" }

// without endDate (open-ended)
"scopeValue": { "startDate": "2025-01-01", "endDate": null }
```

The metric names and value types are identical to their session counterparts above.

1. Total Routes Completed

POST /api/route-stats/total-routes-completed-by-date-range  
Body: { "startDate": "2025-01-01", "endDate": "2025-12-31" }

```
{
  "success": true,
  "scope": "dateRange",
  "scopeValue": { "startDate": "2025-01-01", "endDate": "2025-12-31" },
  "metric": "totalRoutesCompleted",
  "value": 198
}
```

value : number

2. Total Connected Users

POST /api/route-stats/total-connected-users-by-date-range  
Body: { "startDate": "2025-06-01" }

```
{
  "success": true,
  "scope": "dateRange",
  "scopeValue": { "startDate": "2025-06-01", "endDate": null },
  "metric": "totalConnectedUsers",
  "value": 83
}
```

value : number

### 3. Users Started More Than One Route

```
POST /api/route-stats/users-started-more-than-one-route-by-date-range
Body: { "startDate": "2025-01-01", "endDate": "2025-12-31" }
```

```
{
  "success": true,
  "scope": "dateRange",
  "scopeValue": { "startDate": "2025-01-01", "endDate": "2025-12-31" },
  "metric": "usersStartedMoreThanOneRoute",
  "value": 34
}
```

value : number

---

### 4. Users Completed More Than One Route

```
POST /api/route-stats/users-completed-more-than-one-route-by-date-range
Body: { "startDate": "2025-01-01", "endDate": "2025-12-31" }
```

```
{
  "success": true,
  "scope": "dateRange",
  "scopeValue": { "startDate": "2025-01-01", "endDate": "2025-12-31" },
  "metric": "usersCompletedMoreThanOneRoute",
  "value": 28
}
```

value : number

---

### 5. Users Never Started a Route

```
POST /api/route-stats/users-never-started-route-by-date-range
Body: { "startDate": "2025-01-01", "endDate": "2025-12-31" }
```

```
{
  "success": true,
  "scope": "dateRange",
  "scopeValue": { "startDate": "2025-01-01", "endDate": "2025-12-31" },
  "metric": "usersNeverStartedRoute",
  "value": 11
}
```

value : number

---

### 6. Users Started But Not Finished

```
POST /api/route-stats/users-started-but-not-finished-by-date-range
Body: { "startDate": "2025-01-01", "endDate": "2025-12-31" }
```

```
{
  "success": true,
  "scope": "dateRange",
  "scopeValue": { "startDate": "2025-01-01", "endDate": "2025-12-31" },
  "metric": "usersStartedButNotFinished",
  "value": 9
}
```

value : number

---

## 7. Total Replays

```
POST /api/route-stats/total-replays-by-date-range
Body: { "startDate": "2025-01-01", "endDate": "2025-12-31" }
```

```
{
  "success": true,
  "scope": "dateRange",
  "scopeValue": { "startDate": "2025-01-01", "endDate": "2025-12-31" },
  "metric": "totalReplays",
  "value": 47
}
```

value : number

---

## 8. Avg Routes Per User

```
POST /api/route-stats/avg-routes-per-user-by-date-range
Body: { "startDate": "2025-01-01", "endDate": "2025-12-31" }
```

```
{
  "success": true,
  "scope": "dateRange",
  "scopeValue": { "startDate": "2025-01-01", "endDate": "2025-12-31" },
  "metric": "avgRoutesPerUser",
  "value": 3.12
}
```

value : number (float, 2 decimal places)

---

## 9. Avg Routes Per Session

```
POST /api/route-stats/avg-routes-per-session-by-date-range
```

Body: { "startDate": "2025-01-01", "endDate": "2025-12-31" }

```
{
  "success": true,
  "scope": "dateRange",
  "scopeValue": { "startDate": "2025-01-01", "endDate": "2025-12-31" },
  "metric": "avgRoutesPerSession",
  "value": 49.5
}
```

value : number (float, 2 decimal places)

## 10. Replay Ratio

POST /api/route-stats/replay-ratio-by-date-range

Body: { "startDate": "2025-01-01", "endDate": "2025-12-31" }

```
{
  "success": true,
  "scope": "dateRange",
  "scopeValue": { "startDate": "2025-01-01", "endDate": "2025-12-31" },
  "metric": "replayRatio",
  "value": {
    "byPlayer": 0.3125,
    "byRoute": 0.0952
  }
}
```

value : **object** { byPlayer: number, byRoute: number } — same structure as session version.

## Quick Reference Table

| Metric                         | Session endpoint suffix                        | Date-range endpoint suffix                        | value type |
|--------------------------------|------------------------------------------------|---------------------------------------------------|------------|
| totalRoutesCompleted           | total-routes-completed-by-session              | total-routes-completed-by-date-range              | number     |
| totalConnectedUsers            | total-connected-users-by-session               | total-connected-users-by-date-range               | number     |
| usersStartedMoreThanOneRoute   | users-started-more-than-one-route-by-session   | users-started-more-than-one-route-by-date-range   | number     |
| usersCompletedMoreThanOneRoute | users-completed-more-than-one-route-by-session | users-completed-more-than-one-route-by-date-range | number     |



|                            |                                           |                                              |                             |
|----------------------------|-------------------------------------------|----------------------------------------------|-----------------------------|
| usersNeverStartedRoute     | users-never-started-route-by-session      | users-never-started-route-by-date-range      | number                      |
| usersStartedButNotFinished | users-started-but-not-finished-by-session | users-started-but-not-finished-by-date-range | number                      |
| totalReplays               | total-replays-by-session                  | total-replays-by-date-range                  | number                      |
| avgRoutesPerUser           | avg-routes-per-user-by-session            | avg-routes-per-user-by-date-range            | number (2dp)                |
| avgRoutesPerSession        | avg-routes-per-session-by-session         | avg-routes-per-session-by-date-range         | number (2dp)                |
| replayRatio                | replay-ratio-by-session                   | replay-ratio-by-date-range                   | { byPlayer, byRoute } (4dp) |

All endpoint paths are prefixed with `/api/route-stats/`.